



Little Wonderer set a sheet of paper beside a tub. "Could this carry a penny?" Big Wonderer put down a short list of folding commands. Little traced the first line. The paper was flat and flimsy. Somehow, those steps would have to give it a new shape.



"Fold the paper in half. Fold the corners down." Little followed each command in order. More folds made an open boat with walls and a broad bottom. "Like a program," Little said, "each step helps the next one work." Now came the test. Would the folds hold?



Little lowered the boat into the tub. It floated: it stayed at the water's surface. The boat pushed water aside, or displaced it. Water pushed back up. That push equaled the weight of the displaced water. When the upward push balanced the boat's weight, the boat stopped sinking lower.



Little added a penny, then another. This load was the weight the boat carried. The boat sank lower, displacing more water. Its waterline, where water met the paper, climbed up the sides. Folded walls kept water out. The broad bottom and air-filled space inside helped the boat displace enough water.



"Add one penny. Watch the waterline," Little wrote. Then one more penny made the rim dip under. Water spilled inside. The boat was swamped! A flat sheet had no walls to protect an air-filled space. Shape mattered, and even this folded boat could not carry an endless load.



Later, the boat sagged. Paper's tiny plant fibers absorb water: they take it in. Wet paper grows heavy and soft until its folds lose their shape. A crayon or wax coat slows soaking. Little saved the commands. "Could my next boat last longer?"

Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Pictures are simplified and not to scale. Look closely, wonder, and pause whenever you like.

Your turn

Write one command sentence for Little's boat. Tell what to do.

Example: First fold the paper in half, then fold the corners down.

WONDER DAILY

What holds a paper boat?

What holds a paper boat?



One leaf. Six steps. A story that keeps going.

GRADE 3